

Animal Recognition System — Project Syllabus

Wildlife identification in constrained environments using dual-camera visible/IR image capture

Term: August 2026 → | **Development site:** California, USA | **Deployment site:** Uganda

Mission Statement

Build a biodiversity-monitoring camera system for heavily wooded, low-light wilderness that recognizes animals from **partial, occluded, motion-blurred imagery** by fusing simultaneous visible-light photographs with aligned infrared (IR) heatmaps. Develop and validate on North American species (deer, mountain lion, coyote, fox, bear), then domain-adapt to Ugandan wildlife (gorillas and other indigenous species).

Source Documents

This syllabus consolidates:

1. The high-level project definition (Block 1 of `Project_Plan_2026_August.ipynb`)
 2. The "Useful Q&A with Gemini" research thread (Blocks 3–8 of the same notebook)
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Part I — Adopted Architecture Decision

The Recommendation We Are Incorporating

Of the recommendations in the Gemini Q&A, the one that best matches the choices already made in the project definition (Block 1) is:

A two-stage hybrid pipeline — MegaDetector (CNN/YOLO) as detection frontend → BioCLIP (ViT) as species classifier — with the IR heatmap injected via *Early Fusion* (a 4-channel RGB+IR patch-embedding layer), training only lightweight heads over a frozen pretrained backbone.

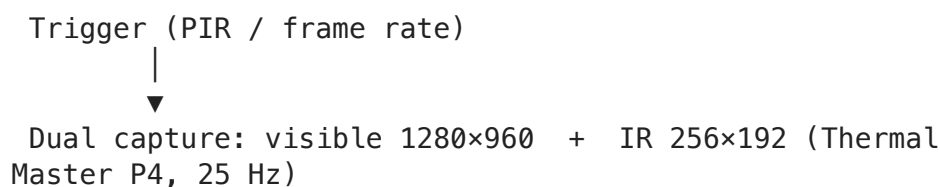
Why this is the best fit for Block 1's choices

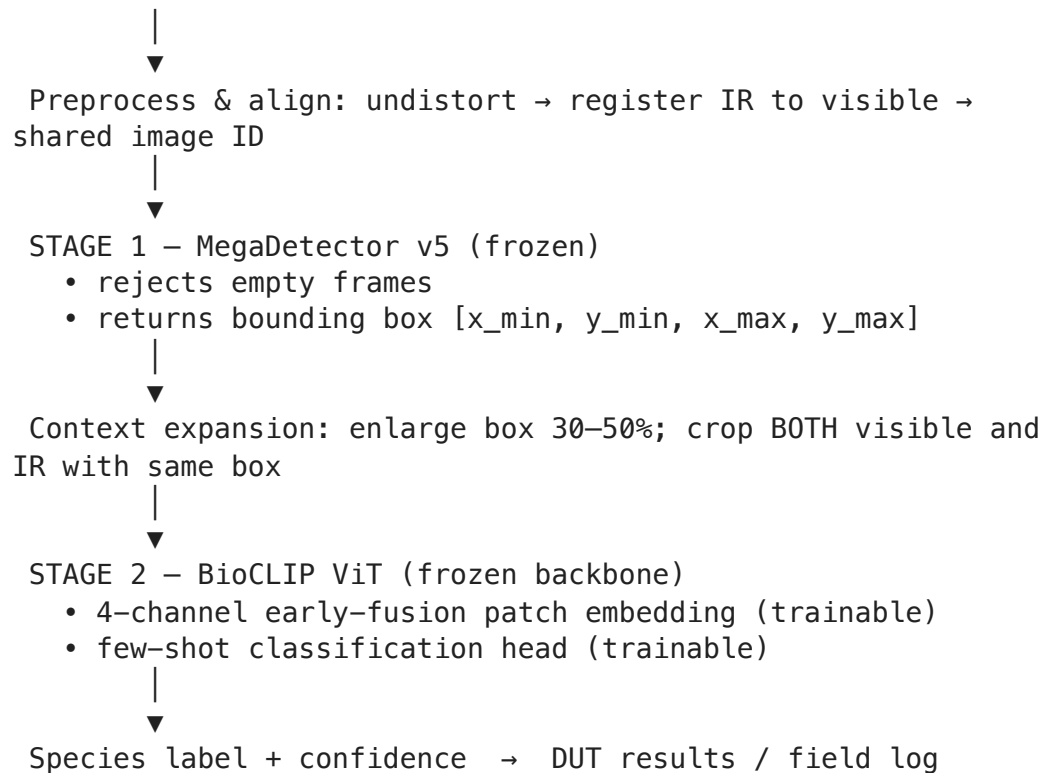
Block 1 Choice	Matching Gemini Recommendation	Rationale
Frontend filter = MegaDetector v5	"MegaDetector as a pre-processor for a downstream classification layer is the exact gold standard"	Pretrained on millions of camera-trap images; already robust to occlusion, low light, and motion blur — directly solves the <i>partial-image deficit</i> out of the box.
Classifier = BioCLIP (ViT)	BioCLIP = "Top Recommendation" among downstream ViTs	ViT-Base backbone with contrastive language-image pretraining on iNaturalist; covers both North American mammals and African primates — ideal for the CA → Uganda migration.
"Modify ViT model to accept IR layer with RGB image data"	Strategy A: Early Fusion (Input Adaptation) — expand the patch-embedding layer from 3 → 4 channels (R, G, B + IR); rated ★★★★★ for BioCLIP	Simplest viable fusion; only the new input embedding + classification head are trained, so a few-hundred-image dataset does not overfit. BioCLIP can still map fused features to its text vocabulary ("gorilla", "coyote") with very little data.
Small starting dataset (hundreds of images)	"Freeze the backbone... your few hundred images only train a small fraction of parameters"	Pure ViTs trained/heavily fine-tuned on small data overfit immediately (no inductive bias). Frozen-backbone few-shot heads neutralize the data-scarcity wall.
Target hardware = Raspberry Pi 5 8GB + Hailo-8 26T AI HAT	Gemini's own hardware verdict: "deploy using a Raspberry Pi 5 paired with a Hailo accelerator — best balance of field durability, low power draw, and high-speed local processing"	The Q&A independently converged on the hardware already chosen in Block 1. YOLO-family MegaDetector compiles cleanly to a Hailo <code>.hef</code> ; 5–7 W power budget suits solar/battery field deployment.

Recorded fallback (do not implement yet)

If early-fusion accuracy stalls on heavily occluded subjects, the designated fallback is **Strategy B: Late Fusion** — dual frozen backbones (RGB stream + IR-as-grayscale stream) joined by a small trainable cross-attention bridge, with **DINOv2** (rated ★★★★★ for this strategy) replacing BioCLIP. This is deferred because it doubles backbone compute — a real cost on the Hailo-8 — and Block 1 explicitly specifies the single-stream RGB+IR-layer approach.

End-to-end pipeline (adopted)





Part II — Course Modules (Plan of Action)

Each module lists **objectives**, **tasks**, **deliverables**, and an **exit criterion** that gates the next module. Modules 1–2 can proceed in parallel.

Module 0 — Project Setup & Literature Grounding (*Week 1*)

Objectives: reproducible dev environment; shared understanding of the two foundation models.

Tasks

1. Create the project repo; pin a Python environment (PyTorch, `megadetector` package, `open_clip` / BioCLIP weights, OpenCV, timm).
2. Run stock MegaDetector v5 on sample trail-cam images; confirm bounding-box JSON output.
3. Run stock BioCLIP zero-shot classification on sample RGB crops of the five target CA species.
4. Read the MegaDetector and BioCLIP model cards; note input resolutions, normalization constants, and license terms.

Deliverables: environment lockfile; two smoke-test notebooks (MegaDetector inference, BioCLIP zero-shot).

Exit criterion: both pretrained models produce sensible outputs on public sample images on the dev laptop.

Module 1 — Capture Rig Bring-Up (*Weeks 1–3*)

Objectives: reliable, synchronized visible+IR capture from the Thermal Master P4.

Tasks

1. Bench-test the P4: verify the four dual-vision modes, 1280×960 visible stream, 256×192 VOx IR stream, 25 Hz frame rate, and −20...600 °C range.
2. Decide and document the IR data format (raw temperature array vs. pre-rendered heatmap) — **prefer raw radiometric data**, rendering to a normalized single channel in software so the fusion layer sees consistent input.
3. Build the capture script: timestamp-synchronized visible/IR frame pairs written with a shared image ID.
4. Characterize parallax between the two lenses at typical subject distances (5–30 m).

Deliverables: capture utility; parallax/ `sync` characterization report; ≥100 test frame pairs (any subject).

Exit criterion: frame pairs are captured with < 1-frame timing skew and a documented, repeatable geometric relationship.

Module 2 — Preprocessing & Alignment Pipeline (*Weeks 2–4*)

Objectives: implement Block 1's preprocessing steps as a reusable library.

Tasks

1. Calibrate both cameras (checkerboard for visible; heated calibration target for IR); compute the homography that maps the 256×192 IR frame onto visible coordinates.
2. Implement alignment: undistort → warp IR to visible frame → upsample IR to crop resolution (bilinear; preserve relative temperature values).
3. Implement the fixed-border animal isolation crop and the 30–50% context expansion around MegaDetector boxes, applied identically to both modalities.
4. Emit paired, aligned, ID-tagged crops in a documented dataset layout.

Deliverables: `preprocess/` library with unit tests; alignment error report (target: mean registration error ≤ ~2 visible-pixels at working distance).

Exit criterion: given a raw frame pair, the pipeline emits an aligned 4-channel (RGB+IR) crop with one command.

Module 3 — Dataset Collection & Tagging (Weeks 3–8, ongoing)

Objectives: several hundred tagged, paired crops of CA wildlife.

Tasks

1. Field-deploy the rig in wooded CA sites; collect deer, coyote, fox, mountain lion, and bear encounters (expect deer to dominate — plan for class imbalance).
2. Run MegaDetector over all captures to pre-filter blanks; manually verify boxes and tag species (single shared ID per visible/IR pair, per Block 1).
3. Split data by *encounter/site*, not by frame, into train/val/test (~70/15/15) to prevent near-duplicate leakage.
4. Log per-image conditions (light level, occlusion severity, motion blur) to enable stratified evaluation later.

Deliverables: versioned dataset v1 (target ≥ 300 –500 tagged pairs); labeling guide; class-balance report.

Exit criterion: dataset v1 frozen with clean splits and ≥ 5 examples per species in every split (fall back to merged "rare-carnivore" classes if bears/lions are too sparse).

Module 4 — Stage-1 Frontend Integration (Weeks 5–6)

Objectives: MegaDetector wired in as the production frame filter and cropper.

Tasks

1. Wrap MegaDetector v5 inference behind a stable interface (`detect(frame) -> boxes, confidences`).
2. Tune the confidence threshold on val data for the *frame-filter* task: maximize blank rejection while keeping animal-frame recall $\geq 95\%$.
3. Implement the dual-modality crop step (same expanded box applied to RGB and aligned IR).
4. Evaluate frontend-only performance on the occlusion-stratified val set; this is the baseline the fusion classifier must beat.

Deliverables: frontend service module; threshold-tuning report; baseline detection metrics.

Exit criterion: $\geq 95\%$ of animal-containing val frames pass the filter with a usable box.

Module 5 — Stage-2 Classifier: BioCLIP + Early Fusion (Weeks 6–10) ← core of the project

Objectives: the adopted recommendation — 4-channel early-fusion BioCLIP with frozen backbone.

Tasks

1. **Zero-shot baseline (RGB-only):** run frozen BioCLIP zero-shot on val crops. Record accuracy — this is the floor.
2. **Linear-probe baseline (RGB-only):** train only a classification head on RGB crops. This isolates the marginal value of IR later.
3. **Early-fusion surgery:** expand BioCLIP's patch-embedding weight from 3 → 4 input channels. Initialize the RGB channels from pretrained weights; initialize the IR channel randomly (small variance) or as the mean of the RGB channels. Freeze everything else.
4. **Train** the patch-embedding layer + classification head on dataset v1 with the targeted augmentation recipe (Module 6). Use a low LR ($\sim 1e-4$), early stopping on val loss.
5. **(Stretch) contrastive alignment:** add an InfoNCE auxiliary loss pulling paired RGB/IR embeddings together, pushing unmatched pairs apart — per the Q&A, this keeps dark/blurry visible crops anchored to their clean thermal signatures.
6. Compare: zero-shot vs. RGB linear probe vs. RGB+IR early fusion, stratified by light level and occlusion severity. **The IR fusion must show its win specifically on the low-light/high-occlusion strata.**

Deliverables: training code; ablation table (3 configurations × condition strata); model checkpoint v1.

Exit criterion: early-fusion model beats the RGB-only linear probe on overall val accuracy **and** on the low-light stratum. If it does not, escalate to the recorded fallback (DINOv2 late-fusion cross-attention) with a documented decision memo.

Module 6 — Targeted Augmentation & Training Optimization (Weeks 7–10, interleaved with Module 5)

Objectives: simulate the field's failure modes during training (per the Q&A's optimization recipe).

Tasks

1. **Random occlusion** (Cutout/Random Erasing, incl. bark/foilage-textured patches) — forces reliance on partial shapes.
2. **Directional motion blur** on the visible channel — mimics animals in motion.
3. **Low-light/contrast jitter** — severely degrade RGB brightness/contrast **while keeping the IR channel clean**, teaching the model to lean on IR exactly when vision fails.

4. Standard geometric augments (flip, small rotation, scale) applied *identically* to RGB and IR to preserve alignment.
5. Address class imbalance: weighted sampling or focal loss for rare carnivores.
6. **(Stretch, only if data grows to thousands)** multi-task auxiliary box/mask prediction head, per the Q&A's multi-task learning suggestion.

Deliverables: augmentation module with visual sanity-check notebook; before/after ablation rows added to the Module 5 table.

Exit criterion: augmentation demonstrably improves val accuracy on the degraded-condition strata without hurting clean-condition accuracy.

Module 7 — Offline End-to-End System Test (Weeks 10–12)

Objectives: Block 1 step 7 — validate the full pipeline in offline mode on held-out data.

Tasks

1. Assemble the full chain (capture files → preprocess/align → MegaDetector → crop → fused BioCLIP → label) as one offline batch job.
2. Run on the untouched test split; report per-species precision/recall/F1, confusion matrix, and end-to-end latency per frame pair on the laptop.
3. Failure review: manually inspect every misclassification; categorize (alignment error, detector miss, fusion failure, label noise).
4. Fix the top failure category; re-run once. Freeze **System v1**.

Deliverables: end-to-end evaluation report; failure-mode taxonomy; frozen System v1 tag.

Exit criterion: documented test-set metrics with no unexplained failure clusters; go/no-go decision for edge porting.

Module 8 — Edge Deployment: Raspberry Pi 5 + Hailo-8 (Weeks 12–16)

Objectives: port System v1 to the target hardware from Block 1.

Tasks

1. Export MegaDetector (YOLOv5) to ONNX → compile to a Hailo `.hef` with the Hailo Dataflow Compiler; validate detection parity vs. the laptop (mAP delta budget ≤ 2 points).
2. Quantize (INT8) using a calibration set drawn from dataset v1 (must include low-light images so quantization ranges cover them).

3. Port the fused BioCLIP classifier: attempt Hailo compilation; if ViT ops are unsupported, run Stage 2 on the Pi 5 CPU (acceptable — Stage 2 fires only on detector hits) and record throughput.
4. Rebuild the capture + preprocess chain on the Pi (P4 camera interface, alignment warp).
5. Measure the full power profile: idle / PIR-wait / active-inference; extrapolate solar+battery runtime for the Uganda duty cycle.
6. Burn-in: multi-day unattended field run in CA; compare edge outputs against laptop outputs on the same frames.

Deliverables: Hailo build scripts; parity report (accuracy laptop vs. edge); power budget document; field burn-in log.

Exit criterion: edge system matches laptop accuracy within budget and meets the field power envelope.

Module 9 — Domain Adaptation: California → Uganda (*future term*)

Objectives: the Q&A's Phase-2 migration plan.

Tasks

1. Keep the frozen MegaDetector backbone and the trained fusion embedding **unchanged**.
2. Swap only the final classification head; retrain on a few hundred Ugandan crops (gorillas first — exploit BioCLIP's existing primate coverage and zero-shot capability for a head start).
3. Re-validate alignment/calibration on the deployed rigs (temperature range and vegetation differ).
4. Repeat Module 7's offline evaluation on Ugandan data before live monitoring.

Deliverables: Uganda classification head; adaptation evaluation report.

Exit criterion: gorilla/indigenous-species accuracy comparable to CA-phase results with $\leq$ a few hundred new labels.

Part III — Milestone Schedule & Assessment

Milestone summary

#	Milestone	Target week	Gate
M0	Both pretrained models running locally	1	Module 0 exit
M1	Synchronized visible/IR capture proven	3	Module 1 exit
M2	One-command aligned 4-channel crops	4	Module 2 exit
M3	Dataset v1 frozen (≥ 300 tagged pairs)	8	Module 3 exit
M4	Frontend filter $\geq 95\%$ animal recall	6	Module 4 exit
M5	Fusion beats RGB-only baseline	10	Module 5 exit — <i>primary project result</i>
M6	System v1 end-to-end test report	12	Module 7 exit
M7	Pi 5 + Hailo-8 parity & power budget	16	Module 8 exit
M8	Uganda head trained & validated	future	Module 9 exit

Grading rubric (self-assessment)

- **Pass:** M0–M4 complete; RGB-only pipeline classifies CA species end-to-end.
- **Merit:** M5 — measurable IR-fusion gain on low-light/occluded strata (this is the project's novel claim).
- **Distinction:** M6–M7 — the full fused system running on the Pi 5/Hailo-8 within the field power budget.

Standing risks & mitigations

Risk	Likelihood	Mitigation
IR↔visible registration error swamps the fusion signal (256×192 vs 1280×960, parallax)	High	Module 2 error budget + calibration rig; evaluate fusion only after M2 gate passes.
Rare-species data starvation (lions, bears)	High	Merged rare classes; weighted loss; supplement with public camera-trap crops (RGB-only rows in a mixed-modality batch).
Early fusion underperforms	Medium	Pre-approved fallback: DINOv2 late-fusion cross-attention (Part I).
ViT unsupported by Hailo compiler	Medium	Stage 2 on Pi CPU is acceptable (event-driven, low duty cycle); revisit with Hailo SDK updates.
MegaDetector v6 availability/variants shift	Low	v5 is pinned for the term; evaluate v6 compact variants only at Module 8.

Open questions carried forward from the Q&A

1. Dual-camera geometry: is P4's fixed lens pair sufficient, or is a beam-splitter rig needed for tighter registration?
2. IR data format decision (raw radiometric array recommended) — confirm during Module 1.
3. Field power strategy in Uganda (solar vs. lithium packs) — needed before Module 8 power budgeting completes.
4. Frame-rate requirement: PIR-triggered burst vs. continuous 25 Hz — drives the edge duty cycle.

Key references

- MegaDetector — <https://github.com/agentmorris/MegaDetector>
- BioCLIP — <https://github.com/Imageomics/bioclclip-2> · <https://imageomics.github.io/bioclclip/>
- DINOv2 (fallback) — <https://github.com/facebookresearch/dinov2>
- DeepFaune (alternative camera-trap ViT) — <https://www.deepfaune.cnrs.fr/en/>
- Hailo-8 on Raspberry Pi 5 — <https://www.jeffgeerling.com/blog/2026/frigate-with-hailo-for-object-detection-on-a-raspberry-pi/>
- Full citation list: see the Q&A blocks in `Project_Plan_2026_August.ipynb`